

DETERMINATION OF COEFFICIENT OF DISCHARGE AND CALIBRATION OF A VENTURIMETER

AIM:-

1. To determine the coefficient of discharge of the given venturimeter and plot the graph Q_a Vs H_m
2. To calibrate the Venturimeter.

APPARATUS USED

1. Venturimeter fitted along a pipe.
2. Differential 'U' tube mercury column manometer.
3. Collecting tank with a piezometer.
4. Stop watch.

PRINCIPLE

The coefficient of venturimeter is the ratio between actual discharge and theoretical discharge,

$$C_d = \frac{Q_a}{Q_{th}}$$

Actual discharge,

$$Q_a = \frac{A \times R}{t} \text{ cm}^3/\text{s}$$

Where

A = Internal plan area of collecting tank. in cm.

R = Rise of water collected in cm.

t = Time for collection in 'R' rise in second.

Theoretical discharge,

$$Q_{th} = \frac{a_1 a_2 \sqrt{2 g H_w}}{\sqrt{(a_1^2 - a_2^2)}} \text{ cm}^3/\text{sec}$$

OBSERVATIONS AND TABULATION

Sl.No	Manometer readings				Time for 'R' cm rise, t sec	Actual discharge Q_a cm ³ /s	Theoretical discharge Q_{th} cm ³ /s	Coefficient of discharge, C_d
	Left limb h_1 cm of Hg	Right limb h_2 cm of Hg	$H_m = h_2 - h_1$ cm of Hg	$H_w = H_m \times 12.6$ cm of water				
Unit	cm of Hg	cm of Hg	cm of Hg	cm of water	sec	cm ³ /s	cm ³ /s	
1								
2								
3								
4								
5								
6								

Where,

$$\begin{aligned} a_1 &= \text{Area of inlet} &= \pi/4 d_1^2 \text{ in cm}^2, \text{ where } d_1 \text{ the diameter of inlet.} \\ a_2 &= \text{Area of throat} &= \pi/4 d_2^2 \text{ in cm}^2, \text{ where } d_2 \text{ the diameter of throat in cm.} \\ g & &= \text{Acceleration due to gravity in cm/s}^2 \end{aligned}$$

H_w = venturi head difference in terms of flowing liquid

$$= H_m \times \frac{(S_m - S_1)}{S_1} \text{ in cm of water}$$

Where $H_m = h_1 - h_2$

$$\begin{aligned} h_1 &= \text{left limb manometer reading in cm of Hg} \\ h_2 &= \text{Right limb manometer reading in cm of Hg} \\ S_m &= \text{Specific gravity of manometer liquid (Hg)} \approx 13.6 \\ S_1 &= \text{Specific gravity of flowing liquid, (Water)} = 1 \end{aligned}$$

PROCEDURE

1. The diameters of inlet and throat are recorded
2. The internal plan dimensions of the collecting tank are measured
3. Open the out let vale of the venturimeter and close the bypass valve
4. Start the pump and water is allowed to flow through the venture meter
5. The inlet valve of the manometer is fully opened and the manometric heads in both limbs (h_1 & h_2) are noted
6. The out let valve of the collecting tank is closed and the time 't' for 'R' rise of the water in the collecting tank is noted
7. The above procedure is repeated by gradually increasing the flow. By partly close the bypass valve. (Minimum 6 sets)
8. The observations are tabulated and the coefficient of discharge is calculated

SAMPLE CALCULATION { SET No..... }

Area of collecting tank, A	=	cm ²
Rise of water, R	=	cm
Time for collection, t	=	sec
Actual discharge, $Q_a = \frac{A \times R}{t}$	=	cm ³ /sec
Diameter of inlet, d ₁	=	cm
Area of inlet, a ₁	=	cm ²
Diameter of throat, d ₂	=	cm
Area of throat, a ₂	=	cm ²
Acceleration due to gravity, g	=	cm/sec ²
Left limb manometer reading, h ₁	=	cm
Right limb manometer reading, h ₂	=	cm
Specific gravity of manometric liquid, S _m	=	13.6
Specific gravity of flowing liquid, S ₁	=	1
Venturi head, H _w = (h ₁ - h ₂) x 12.6	=	cm
Theoretical discharge, $Q_t = \frac{a_1 a_2 \sqrt{2 g H_w}}{\sqrt{a_1^2 - a_2^2}}$	=	cm ³ /sec.
Coefficient of discharge, C _d	=	

CALIBRATION

Actual discharge through the ~~VENTURIMETER~~ is assumed to be governed by the equation,

$$Q_a = K H_w^n$$

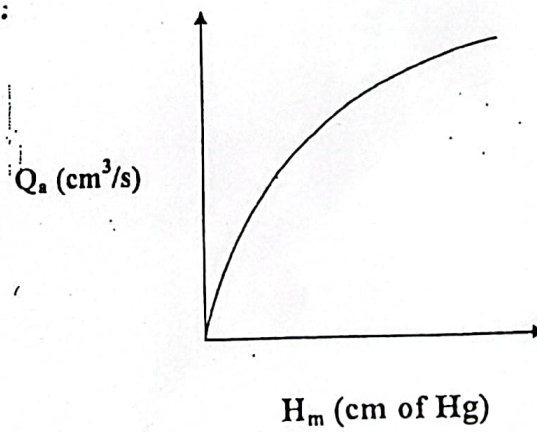
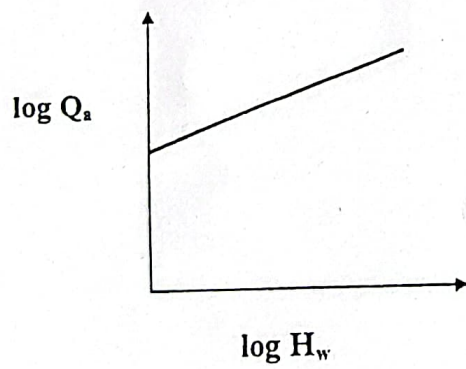
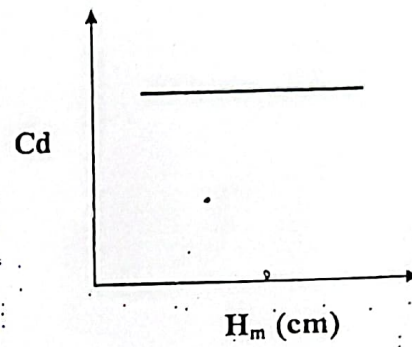
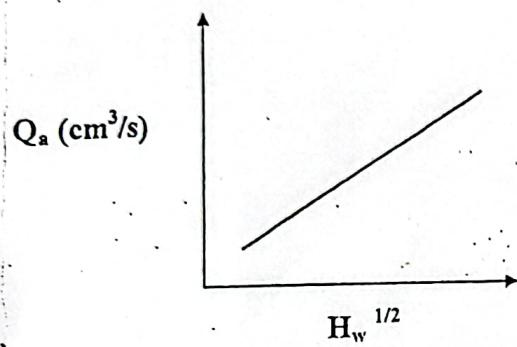
Where

Q_a = Actual discharge

H_w = Pressure difference between the inlet and the throat (cm of water)

K and n are constants

Model graphs for Venturimeter



Taking log on both sides

$$\log Q_a = \log K + n \log H_w$$

Plotting $\log Q_a$ Vs $\log H_w$ graph, the slope of the line gives the value of n and the intercept on the $\log Q_a$ axis gives the $\log K$, Substituting these values of K and n in the equation for discharge $Q_a = KH_w^n$, the calibration equation can be obtained.

Q_a can be calculated for suitable values of H_m (converted to H_w). The calibration curve can be plotted with H_m on X-axis and Q_a on Y-axis.

RESULT

Coefficient of discharge of the venturimeter is, C_d =

Actual discharge for a pressure difference of 20 cm of Hg = cm^3/sec

INFERENCE

GRAPHS

1. C_d Vs H_m
2. Q_a Vs H_m
3. $\log Q_a$ Vs $\log H_w$
4. Calibration Graph, Q_a Vs H_m